

A Facility Design for Wiring.com. A new Wire Net Manufacturing Plant in Sedibeng District

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1. Executive summary

A team of industrial engineers has been appointed as facility designers to design a new facility for Wiring.com. The facility is a green facility and therefore needs to be designed from scratch. The factory is required to make 4 different types of netting with 4 different processes.

1.1. Key Requirements

Key requirements outline what is critical to the customer for the design. For the design provided the objective is to meet the design. Consideration for each requirement is weighed during the alternative and master plan for the facility. The key requirements identified for wiring.com are:

- Must minimize the space required.
- Lean manufacturing for efficient operation
- Minimize the operational costs associated with inefficient facility layout.
- Meet the client demand with sufficient production capacity.
- Deliver a facility plan to be implemented on the new site.

The design approaches used by the engineering team ensures requirements are met in the final design.

1.2. Design approach

A research process and data overview on the process is concluded. The overview includes analysis on the different equipment and relationships linked. Different alternatives were evaluated before conducting the final design. Each member of the design team is tasked with producing an unbiased alternative. The designs are noted in section 7. The designs are then evaluated based on a criterion. The criteria links to the objects and requirements by the client. From the alternatives a final design is created with key design considerations and a business case presented.

1.3. Key design considerations

The facility layout is planned to ensure a continuous workflow of materials and reduce the risks of bottlenecks. Workstations were set close to each other in an assembly line design to ensure that the material flowed in a sequential way from one process to the following. Different considerations are weighed up for the overall layout including the transportation and delivering of raw materials and the collection of smaller orders. All nestled together to form an integrated supply chain for wiring.com.

2. Background

The manufacturing company specializing in wire-based products, Wiring.com, operates within the steel and wire industry. The steel and wire industry serves a wide market, including construction, automotive, manufacturing, and aerospace. The steel and wire market are experiencing rapid growth in recent times.

The steel wire market is influenced by a variety of factors that influence the market's trajectory. Important factors include changes in consumer preferences, technological advancements, and raw material prices.

Wire mesh's usages include rectangular nettings used for property enclosures, while hexagonal nettings are employed in agriculture and forestry to enclose woodland plantations, protecting them from animal intrusion. Additionally, these nettings serve as slope reinforcement, defending against rockslides. Furthermore, a specialized category of nettings, known as round braids, serves as reinforcements for hoses and cables or shields cables against electromagnetic interference.

To captivate sufficient market share in the competitive but growing market pressure is applied on manufacturing facilities. Emphasis is placed on deliverability and production capacities. This requires consideration of multiple constraints relevant to the wire mesh or netting industry. Targets are set on maximizing profits using lean manufacturing and increased production efficiency.

Our client Wiring.com requires a bigger facility for the manufacturing of 4 different products given the market pressure. These products include bird netting, chicken netting, jackal netting, and pig netting. The facility should be designed from the conceptual design phase as a new manufacturing plant. The plant will focus on the relevant market requirements.

3. Design Problem and Objectives

Wiring.com needs to construct a new manufacturing facility on a 100 HA site in the Sedibeng District, Gauteng, South Africa, due to a rapid increase in demand for netting products. The facility must be designed from the conceptual phase to accommodate lean manufacturing principles and industrial engineering concepts to ensure operational efficiency. Since a new manufacturing facility have to be designed, numerous considerations arise. These include determining an optimal layout and space allocation within the allocated site to accommodate manufacturing operations, administrative offices, and supporting facilities. Additionally, prioritising adequate parking space, implementing security measures, ensuring regulatory compliance, and managing costs effectively are crucial components of the design process.

3.1. Problem Origin

Wiring.com recently experienced a rapid increase in demand for netting products, necessitating more production capacity. This resulted in Wireing.com purchasing a new site for development to be able to expand their production capacity.

3.2. Problem Statement

Wiring.com's current facility cannot meet the manufacturing requirements. The purchase of this new site and the desire to increase production capacity necessitates the need for a facility design for a new manufacturing facility.

3.3. Objectives:

- Determine space requirements activities.
- Design an efficient manufacturing facility.
- Incorporate lean manufacturing principles.
- Ensure the new facility has the capacity to meet the market demand and requirements set out specification document.
- Ensure the facility complies with regulatory requirements.
- Design an effective packaging process.

3.4. Deliverables

- Daily production plans.
- Process flow charts showing the process of manufacturing the desired products.
- Detailed layout design of the manufacturing facility, including floor plans, equipment placement, and infrastructure layout.
- Alternative design layouts.
- Evaluation matrix comparing different layout designs.
- Capacity analysis demonstrating the facilities ability to meet market demand.

4. Requirements and Criteria

Table 1 Requirements Document

Plant Specification Outline	
1.	Facility Requirements
1.1 Purpose	This document sets out the specifications and requirements for the design and development of a facility for Wiring.com
1.2 Scope	The document covers the physical, functional, and performance requirements of the facility design.
2.	Physical Requirements
2.1 Location	• 10Km from the current facility • Sedibeng District in Gauteng, South Africa
2.2 Size	• 100HA = 1 000 000 meter squared • The shape of facility does not have requirements. • Space minimized to reduce costs
2.3 Infrastructure	The current plant has no infrastructure and is a greenfield design.
2.3.1 Parking	Sufficient parking for all staff assumes 1 parking per staff member and minimum 6 extra parking for visitors etc.
2.3.2 Lockers	1 locker per staff member
2.3.3 Rest Rooms	Rest Rooms in each department and for both genders (2 categories)
2.3.4 Offices	1. Plant management: 1 office per manager. 2. Operations department: 3. Sales: 1 open floor office for sales team 4. Purchasing: 1 office to accommodate procurement team 5. Finance: 1 office per person

	6. Inventory management: Inventory management share 1 office with procurement. 7. Admin 8. Reception offices: 1 open plan desk at entrance of office with reception area for customer seating 9. Maintenance and spare part storage rooms: room sufficient to store parts and safety equipment. 10. All supervisors: open plan office 1 desk per person	
2.3.5 Delivery Lot	Collection bay for trucks and customers collection sufficient to handle demand	
2.3.6 Support Systems	Lighting, ventilation, electrical, plumbing, and safety	
3.	Product Requirements	
3.1 Products	Wire Netting 1. Bird Netting 2. Chicken Netting 3. Jackal Netting 4. Pig Netting All products are galvanized with light finish	
3.1.1 Bird Netting	Roll Height (m)	0.9
	Aperture (mm)	13
	Wire Diameter (mm)	0.71
	Roll length (m)	50
	Weight (kg) per roll	22.5
3.1.2 Chicken Netting	Roll Height (m)	1.8
	Aperture (mm)	25
	Wire Diameter (mm)	0.9

	Roll length (m)	50
	Weight (kg) per roll	41.5
3.1.3 Jackal Netting	Roll Height (m)	0.6
	Aperture (mm)	75
	Wire Diameter (mm)	1.8
	Roll length (m)	50
	Weight (kg) per roll	19.5
3.1.4 Pig Netting	Roll Height (m)	0.9
	Aperture (mm)	90
	Wire Diameter (mm)	1.8
	Roll length (m)	30
	Weight (kg) per roll	14.7
3.2 Methodology	• Lean manufacturing • Profit maximization	
3.3 Raw Materials	Rods are bought from a supplier. Galvanization-zinc bought from an external supplier	
3.3.1 Mild steel round bars	• 5mm diameter • 3m length bars • Cost: R42.40/3m-bar	
3.3.2 Galvanising	• Form: Zinc ingots • Cost: R40000/ton	
3.3.3 Raw Material Operations	• Drawing Rods to spec diameter • Galvanisation • Net Production	

4.	Labour
4.1 Normal Labour Availability	220 days/annum - Assume 8 production hours per day
4.2 Abnormal Labour Availability	Overtime Labour to be specified in future
5.	Space Requirements
5.1 Storage Space	Storage for 4% scrape rate during the first operation.
5.2 Outbound Inventory	As lean as possible since the <u>stock is sold as it is made.</u>
6.	Customer Requirements
6.1 Products Demand	- Bird Netting = 6250 tons/year - Chicken Netting = 6250 tons/year - Jackal Netting = 2500 tons/year - Pig Netting = 10 000 tons/year Total: 25 000 tons/year
6.1.1 Demand in rolls	Formula: $(\text{demand}[\text{tons}]) \times 1000 / (\text{roll weight}[\text{kg}]) = \text{roll unit demand/annum}$ (rounded up to nearest unit) For calculations see Appendix B: Demand and Throughput Data. - Bird Netting = 277778 rolls - Chicken Netting = 149165 rolls - Jackal Netting = 128206 rolls - Pig Netting = 680273 rolls
6.1.2 Demand Round Bars	255 587 units per day including 4% waste. For calculations see Appendix B: Demand and Throughput Data.
6.1.3 Demand In wire	<u>Yearly per Netting Type</u>

	Bird [0.7mm]: 2 011 981 608 m Chicken [0.9mm]: 1 252 148 060 m Jackal [1.8mm]: 125 214 806 m Pig [1.8mm]: 500 859 224 m <u>Daily Demand per Diameter</u> 0.7mm: 9 145 371 m 0.9mm: 5 691 583 m 1.8mm: 2 845 791 m
6.2 Production	Production must meet the demand
6.3 Product specification	• Product meets product specification 3.1.1 to 3.1.4 respectfully. • Product is galvanized. Product packaged for collection or delivery

The requirements set out in the specification document couple the design requirements for Wiring.com's new facility. The requirements will be present in the design of the facility. The specification is used as a parameter and the validation process for the design. Successful facility design will be evaluated, and conclusions drawn on criterion set out in this document.

5. Process, Approach, Assumptions and Constraints

5.1. Preliminary Functional Process Drawing

The visualization of the typical manufacturing process involved with making the netting is required to quantify the approaches and constraints with viable assumptions. Research is done on machines and the process involved. Data for each process and component within the design requires attention and its own set of assumptions and constraints. A preliminary process diagram is concluded and presented in Figure

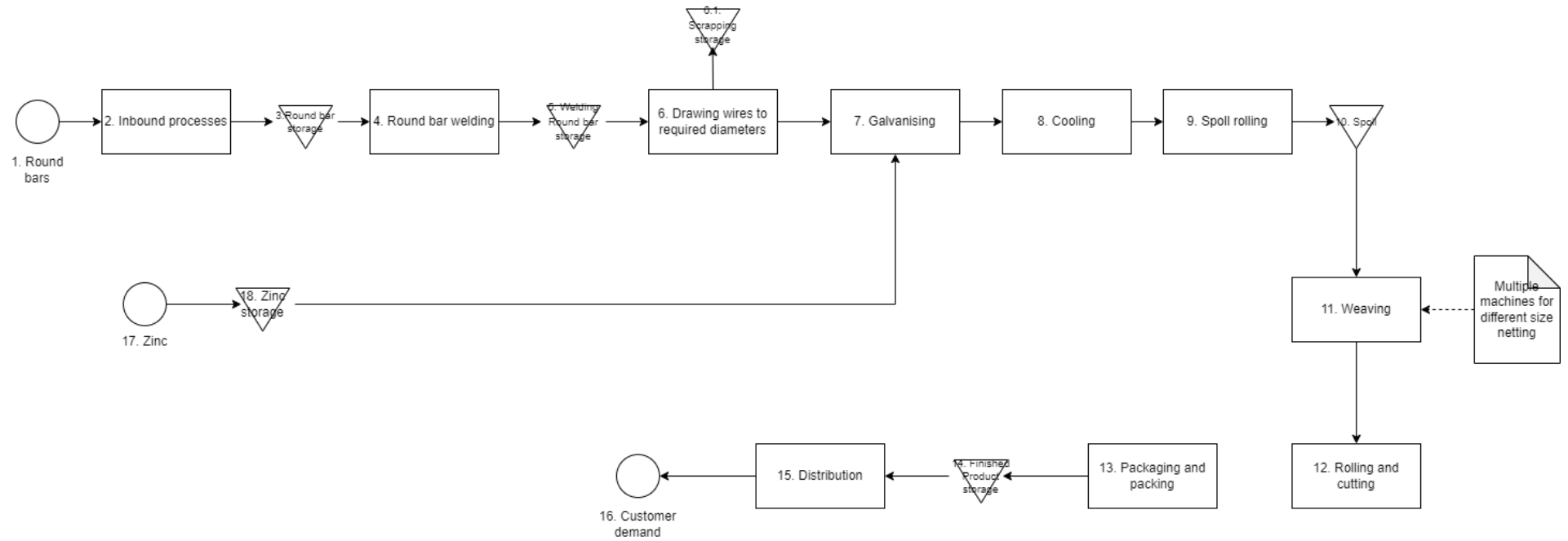


Figure 1: Preliminary Process Diagram for Manufacturing Process

5.2. Data Acquisition Strategy

The data acquisition strategy will outline the method for collecting data to quantify various processes or steps within the design. Due to the constraints of not having a physical site to inspect or walk on the floor a more virtual approach is followed for research. Various sources will be considered to acquire the necessary data as per the preliminary design. Some sources for data collection include.

5.3. Sources

- Online Videos
- Academic Articles
- Web Articles
- Supplier Specification Documents

By use of the set sources sufficient data, constraints, and assumptions can be drawn to give an overview of the data. This will set the building blocks for the green field design of the facility for Wiring.com.

Table 2 The Different Data Required For The Relevant Process

Storage Systems 	• Storage or package method • Amount of storage (space) • Holding period • Component dependencies • Storage component movement.
Processes 	• Process Time (setup, runtime, changeover, etc.) • Process Steps • Process Space Requirements • Process Throughput • Process dependencies (storage and transport to and from process e) • Component dependencies
Material Movement 	• Material movement device (forklift, trolly, labour, etc.)

	• Movement method space (forklift turning and parking etc.) • Component dependencies.
Sources and Sinks	• Supplier delivery rate • Inbound material type (round bars in coil. etc) • Customer demand and interaction

5.4. Assumptions

This section consider all the assumptions required to design each component. Each step in the manufacturing process is numbered and a table is drawn to express the assumptions related to that specific step.

Table 3 Assumptions

Step	Assumption
1.	• Raw material sourced locally. • Order to delivery time 5 working days.
4.	• Assume a 5 mm round bar can be welded/joined before drawing to give continuous flow to the drawing machine. • Assume conservation of material/volume in the drawing process.
6.1.	• Scrap is collected in a bin in the machine, and it can be assumed collection is automated. • Volume is conserved during the drawing process $V_i - Scrap = V_f$ • Machine is assumed to have same drawing speed for all diameter wires given the series nature of drawing dies.
7,8.	• The zinc is fed by hand into a shoot in the machine. • Galvanisation process does not have startup time that influences working hours. Machine is warmed up prior to work hours.

	Galvanization and cooling are done by one machine and one continuous process.
9,10.	Wire is spooled by the galvanization machine
11.	- Assuming the weaving capacity of the hexagonal wire mesh machine allows for the production of multiple rolls of narrower netting, for example, instead of producing a single mesh with a width of 3200 mm, it can produce two meshes, each with a width of 1500 mm. - Assuming the equipment can produce netting with a max width of 3200mm  - Assume that all the spools that are fed into the machine will be replaced at once.
14	Assuming demand are distributed uniformly and that the product will not stay in storage longer than two days.
15.	Assume that there are only two options when it comes to distribution. The netting products can either be collected for small orders or it will be delivered by truck. A small order is defined by any size smaller than a normal full truck load. Assume dedicated loading areas for both collection and delivery.
16.	Assume the customer-required tons per year are distributed uniformly (linear demand).
	Additional Assumptions
	All machines are outputs are judged on a worst-case specification to allow for assumptions on maintenance and downtimes.

More assumptions	R19 = \$1 R20 = 1€
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5.5. Assumption Verification

Evaluating or checking a list of assumptions is critical to the process identification stage. The verification process ensures the building blocks of the design do not bypass any inherent requirements or possibilities. Each assumption is evaluated one by one, seeing if they make sense and if they're true. This entails doing research and asking appropriate questions to see if what is assumed holds up. By doing this any mistakes are identified and corrected within the table. The process makes analysis and decision-making more reliable in the final design.

5.6. Constraints

Similar to assumptions each step in the operation has constraints associated with the step. The constraints are critical to understanding the modification tolerance of each step for the final design. In addition, it explains the interdependencies of components critical to understanding the process capabilities and requirements for each step. With constraints, sufficient knowledge can be gained to modify the current preliminary design of the plant. The results are drawn in a table to and easily distinguished with the step number.

Table 4 Constraints in Process

Step	Constraints
1.	Raw material form factor cannot be changed. The delivery truck size and shape are unknown.
6.	One Machine can only draw one diameter wire at a time and requires setup of 3 min to change dies. Machine must be fed continuous rod stream.
7.	Galvanisation must be done at a rate that exceeds the drawing to avoid the need for intermediate storage.
8.	Cooling operation is a constraint to galvanisation speed.

10.	Storage on galvanized wire is constraint to the speed of output from galvanization process and the input of the weaving operation.
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6. Current State (or) Needs Analysis

6.1. Inbouding Process of Round Bars

Raw materials are delivered in the form of round bars measuring 5 mm in diameter and 3 meters in length. These round bars are unloaded and transported to the storage areas using overhead gantries, as depicted in Figure 2.



Figure 2: Unloading process

Figure 3 illustrates the movement of the round bars to their designated location utilizing an overhead gantry.



Figure 3: Movement to storage

The round bars are subsequently stacked and stored, as demonstrated in Figure 4.



Figure 4: Round bar storage

The calculation for the required amount of mild steel round bars per day is as follows:

$$Weight = volume \times density$$

$$Volume = \pi \times radius^2 \times length$$

Table 5: Round Bar Calculations

Description	Calculated values
Volume of one 3 m Mild Steel Round Bar	$5.89 \times 10^{-5} m^3$
Weight of one 3 m Mild Steel Round Bar	$0.4624 kg$
Total weight required per day	$113\,637 kg$
Considering the 4% waste per day	$118\,183 kg$
Total amount of Mild Steel Round Bar demand per day.	$255\,587 units$

6.2. Round Bar Storage

The formula used to calculate the total volume necessary for storing raw material for one day is as follows:

$$Total\ Volume = volume\ of\ one\ round\ bar \times number\ of\ round\ bars$$

Given that the required space is calculated to be 16 cubic meters for the number of round bars necessary for one day.

6.3. Round Bar Welding

Round bars are assumed to be fed into the drawing machines in continuous lengths. However, the raw material is delivered in 3 m lengths. As a result, a butt-welding operation is used to join two round bars before they enter the drawing machine to ensure continuous feed of raw material.



Figure 5 Butt Welding Machine {Citation}

Figure 5 shows a butt-welding machine. The machine welds two round rods together in a continuous operation. An overview of the machine data is given in Table 6.

Table 6 Butt Welding Data Overview

Butt Welding Process Overview	
Process Time (setup, runtime, changeover, etc.)	3s/round bar as extracted from video source Appendix A3 Welding Equipment.
Process Steps	1. Both ends clamped 2. Machine welds ends. 3. Machine cools joint 4. Machine releases both round bars

Process Space Requirements	5m x 10m meter block that includes space for coiling of joined bars. 
Process Throughput	1m/s
Process dependencies (storage and transport to and from process e)	Process must weld round bars fast enough for input of the drawing machine.
Labour Requirements	Machines require 1 operator per welder.

6.4. Round Bar Coiling and Storage

After the round bars are welded, they are temporarily stored for the drawing process to follow. The round bars are stored in large coils which are then fed into the machine and butt welded to coils that follow to allow continuous drawing operation.

Table 7 Round Bar Coil and Storage Overview

Round Bar Coil and Staging for Drawing	
Storage or package method.	1-ton coils that hold 6500m of coiled 5m round bar 

Amount of storage (space)	1m x 2m Storage and staging area for must allow for two coils before each drawing operation.
Holding period	Holding period changes but in a decoupling point so has continuous movement.
Component dependencies	Coils are staged before drawing operation on de-coiling stands. 
Storage component movement	Coils are moved around on spools that fit onto de-coiling machine and can be transported with forklift.

6.5. Wire Drawing Process

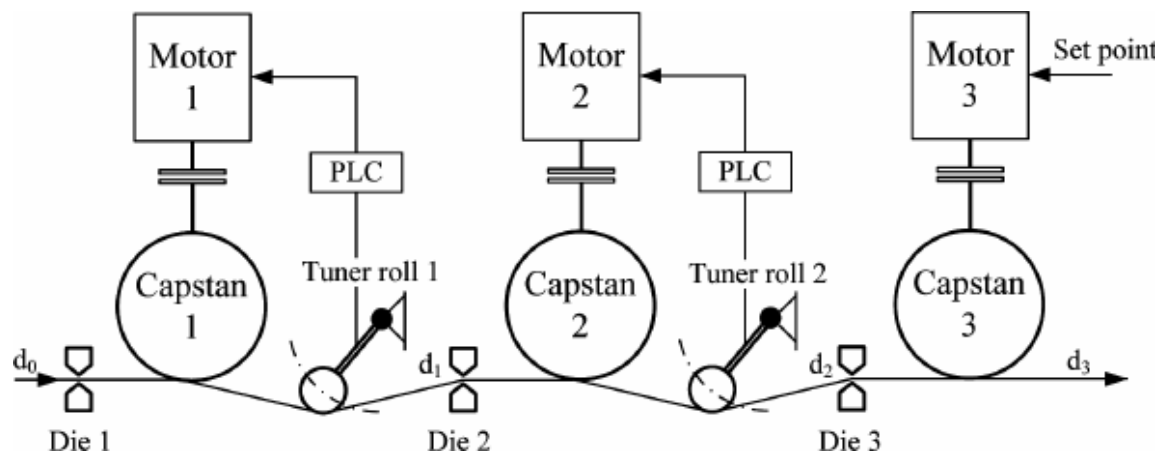


Figure 6 Wire Drawing Process (Tasevski et al., 2015)



Figure 7 Wire Drawing Machine (Tech, 2023)

By looking at Figure 6 and Figure 7 the process and a typical machine is depicted. Figure 6 shows the process with multiple components. The process works by having a feed system in the form of coils as received by our inbound logistics. The system then starts by drawing the wire through a series of dies. The power is supplied by motors attached to the capstans which are basic pullies that the wire rolls around to grip it for the forces present in drawing. The number of dies depends on the reduction required from raw material to the final diameter wire. The final wire can pass through some preparation processes and then be rolled onto a spool for storage.

According to the data acquisition process, the following data is required for a processor. The results of the wiring drawing process are tabulated in Table 8.

Table 8 Wire Drawing Data Overview

Wiring Drawing Process Data Overview	
Supplier 1: DPAU (Data Sheet in Appendix A)	
Process Time (setup, runtime, changeover, etc.)	<u>Setup (setup = number of dies*3.5min)</u> 0.71mm Wire (8-die system) Setup = 8 * 3:30 = 28 min 0.9 mm Wire (8-die system) Setup = 8 * 3:30 = 28 min

	1.8 mm Wire (5-die system) Setup = 5 * 3:30 = 17:30 min <u>Changeover and setup-time are assumed to be the same.</u>
Process Steps	1. Continuous rod feed 2. Drawn through die 1...n (n=number of reductions) 3. The wire is rolled onto the spool in final diameter
Process Space Requirements	The length of the system is 9m and 1.4m per die reduction. the width of the system is approximately 2m. Space for movement around the machine is a 2 m perimeter. Movement space for a forklift to move wire spools for system input and system output is required.
Process Throughput	Wire drawn at 20m/second
Input Resources	Power: 22kw/die
Process dependencies (storage and transport to and from process e)	The system must be fed wire spools as long as possible to avoid system setup requirements.
Labor Requirements	1 Operator/machine

6.6. Scraping and Storage

During the drawing process a 4% scrap is produced. Sufficient space and transport are required for the scrap. Scrap is automatically collected in bins that can be transported by forklift to a storage area. We can assume that the scrap is collected throughout the working day and then removed at end of day.

Table 9: Scraping and Storage Data

Data Description	Data
Storage or package method	Forklift bin 
Amount of storage (space)	1x1m per bin.
Holding period	To be specified.
Component dependencies	The bins are removed daily and do not hinder the process.
Storage component movement	Components are removed with a forklift or forklift hoists. 1 Labourer is responsible for waste removal of all machines.

6.7. Galvanising and cooling



Figure 8 Galvanisation Machine (AMBER wire mesh machine)

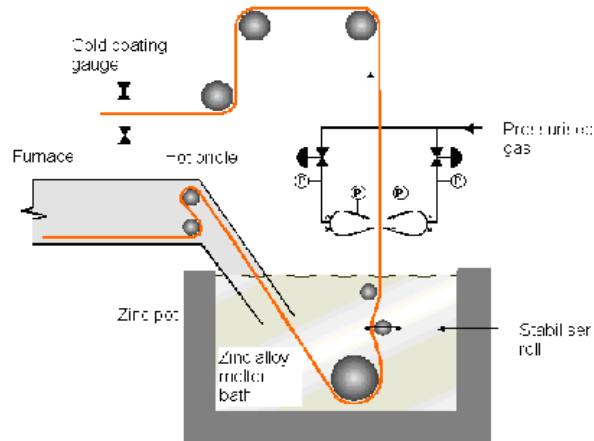


Figure 9 Galvanization Process (Cuzzola & Parisini, 2010)

Figure 8 and Figure 9 shows a typical process and machine for the galvanization of mild steel wires. The machines are supplied with data sheets as per appendix A section A2 which shows the data sheet for an industrial machine supplied by Anber Machine. The data is tabulated according to the data acquisition strategy in Table 10.

Table 10 Galvanization Process Data Overview

Galvanization Process Data Overview	
Process Time (setup, runtime, changeover, etc.)	Changeover is continuous through the process and does not halt the operation of the machine. Runtime assumes 8 hours per day of operation for the machine
Process Steps	Pay off → lead annealing bath → water quench → Fumeless pickle → "Flash" pre-flux → Drying → Hot-dip galvanizing → Wiping → Cooling → Vertical take up
Process Space Requirements	Workhouse Size(W*L): 10m*105m
Process Throughput	30-45 tons of wire/day
Process dependencies (storage and transport to and from process)	The system must be fed wire spools as long as possible to avoid system setup requirements.

Material Input Requirements	Zinc coating is 70-100 g/m ² Power: 50kw
Labour Requirements	2 operators/machine

6.8. Zink Inbound Raw Materials

The galvanisation process takes material input as zinc ingots. The ingots are fed into the machine and coated onto the material. An overview of the material demand handling for raw zinc and the storage of the zinc is required. The table below sets out the requirements in the zinc process.

Table 11 Zinc Raw Material Handling Overview

Zinc Raw Material Data Overview (reference to Appendix B3)	
Supplier delivery rate	1 truck every 13 days with 48 pallets onboard
Inbound material type (round bars in coil. etc)	Ingots
Demand	3680 kg/day of zinc ingots 3.7 pallets/day
Ingot Packaging	Slap Ingot Bundles on pallets 0.45h x 0.88l x 0.44w 1000kg
Supplier delivery process	Truck arrives every 13 days and delivers 48 pallets that is unloaded with a forklift on an unloading station. Pallets are stacked and stored.
Storage Requirements	Storage Surface Required (3 stack height) 7 m ² without forklift movement area

6.9. Wire Spooling and Storage Process

Wire throughout the system as prepared by drawing and galvanisation process are stored and moved from operation to operation. As a result, the wire in various sizes is coiled for storage and movement. Ideally each process would require as little as possible changeover and thus spool sizes are maximized.



Figure 10 Typical Wire Spools (“Big Coil Wire Used as Tie Wire or Binding Wire in Construction”, s.a.)

Figure 10 shows a typical spool that is used for wire storage. This form of storage is ideal since most market machines already use this form of storage as input and outputs the product onto a coil that is strapped and moved. As per the data acquisition method the data for spool storage and specifically process 9 is specified in table.

Table 12 Spooling Storage Data Overview

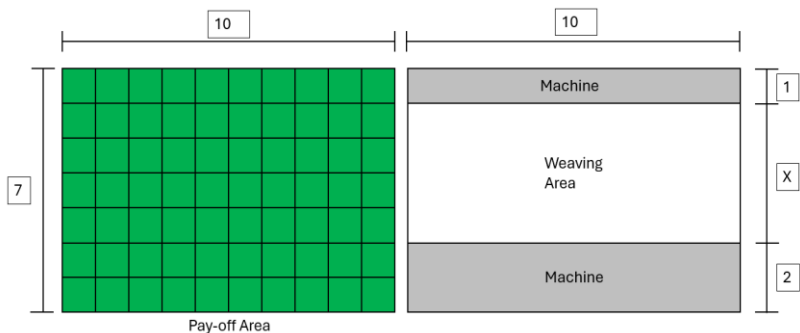
Spooling and Storage of Spools	
Storage or package method	Coils as in figure 6. The coils are stackable.
Amount of storage (process 9)	Each roll required 1x1m for storage. Each roll can store a total of 1 ton. 0.71 mm Wire +-320 000m/spool 0.9 mm Wire (8-die system) +-200 000m/spool 1.8 mm Wire (5-die system) +-50 000m/spool
Holding period	Process 9 is a decoupling point and will only store wire for a short period. Decoupling points are only internal storage.

Component dependencies	Spooling operation is dependent on the galvanization plant. Storage of the system depends on demand from the gabion machines.
Storage component movement	Overhead crane that moves rolls, a forklift is optional if more capacity is required. Trollies can be used to move rolls around from place to place on the floor and to the Gabion machines.
Storage or package method	Stored in a spool that is strapped with nylon tie-downs.

6.10. Weaving Process

Table 13 Weaving Data Overview

Weaving Data Overview	
Process Time (setup, runtime, changeover, etc.)	Normal 8-hour runtime. Assume no changeover time are required to setup the machine with new wire. When the spools are nearly finished new spools are placed below the current spools and the ends are Butt Welded together to ensure a continuous inward flow.
Process Steps	The process of weaving the netting product includes several steps and depends on the equipment used. However, the following steps or components are usually included in this process: The Wire Payoff System: Hold and feeds the raw material into the production process. The Wire Straightening Mechanism: Ensure that the input wire is straightened before fed into the weaving process. Weaving: Forms the hexagonal mesh pattern. Cutting: Cut the netting to the desired length. Rolling or Coiling: Netting is rolled or coiled for easy storage and transport.

	The rolls are cut at the specified roll lengths and automatically rolled onto a conveyer to transport to packaging process. Examples of typical weaving machines in the market as well as their specifications can be seen in Appendix A under the heading A4 Weaving Process.
Process Space Requirements	<u>Weaving, Wire Tension, and Net Rolling Machine</u> Shop floor Width: The space requirements for machines in width are the netting width of 3.2 meters and 2 meters on one side and another 1 meter for machine components. Shop floor length: The machine occupies a total of 10m in length excluding the wire pay off area. Wire Payoff Area: 1x1m per wire winded per machine  Figure Shows Chicken Netting Example
Process Throughput	The process throughput depends on the specific equipment used however on average 100 meters of netting product can be produced in an hour.
Process Dependencies	Process requires wire coils to be transported to the wire payoff system. Process also requires rolls of netting product to be transported to the packaging process. Automatic transportation, using conveyer belts to transport from weaving machine to packaging area.

Labour Requirements	Two labourers required per machine. Responsible to ensue constant wire fed, and general supervision and support. One labourer to always remain with the machine.
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Figure 11 shows a setup of a specific supplier based on their products offered.

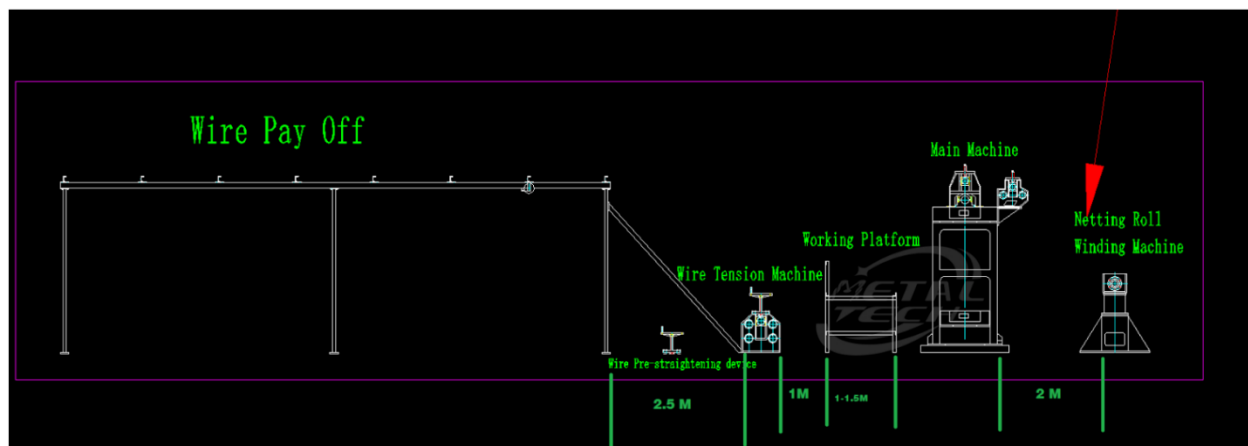


Figure 11 Weaving Machine Diagram

6.11.Packing and Packaging

Table 14: Pet Strapping Data Overview

Pet Strapping Data	
Process Time	The PET strapping is fed automatically from a roll and the only changeover time is the time required to replenish this roll of strapping. This takes about 20 seconds.
Process Steps	Loop Formation: Form a loop of the PET strap around the roll of netting. Tensioning: Use the strapping tool to tension the PET strap around the roll. Sealing: Seal the ends of the strapping together to keep the tension. Cutting: Cut the strap just after the seal.
Process Space Requirements	Machines are hand-held, however take 1.5m x 3m of work area per machine.

Process Throughput	Based on videos watched this process takes an average time of between 10 to 15 seconds per strap. (https://youtu.be/0IAr43z9cUM?si=S3b18rnTIXq8-ZYB)
Process Dependencies	Requires the rolls of netting from the weaving process. Rolls of netting also required to be transported to storage area after strapping.
Labour Requirement	Requires one operator per tool. An example of a typical tool can be seen in Figure 12.



Figure 12 Pet Strapping Machine

Table 15: Stretch Wrapping Data

Stretch Wrapping	
Process Time	Will have changeover time, when replenishing the stretch wrap of the machines. Also requires manual labour to tie the stretch wrap to the bottom corner of the pallet before it can be wrapped.
Process Steps	Palletisation: Place the rolls of netting product on the pallet in organised way as required. Securing: Secure the product to the pallet as necessary with a strap.

	Stretch Wrapping: Use stretch wrapping machines to wrap the entire pallet load tightly. Refer to Figure 5.2. for an example of a packed pallet.
Process Space Requirements	The Typical Machine have the following dimensions: 1595mm x 600mm x 1194mm (L x W x H). Manoeuvrability space is also required for moving pallets to and from the machine.
Process Throughput	20-40 Pallets Per Hour
Process Dependencies	This process requires the pallet to be transported to and from the wrapping machine. Process depends on user to initiate the process.
Labour Requirement	Requires one operator per Machine. An example of a typical stretch wrapping machines in the market as well as their specifications can be seen in Appendix A under the heading A5 Packing And Packaging Process.

Table 16: Product Storage Data

Product Storage After Packing.	
Storage or package method	Products are palletised for ease of storage, packaging and distribution. Pallets are placed on warehouse pallet racks. Pallets are loaded on the rack at one end and collected at the other end, racks contain rollers that moves the pallets forward reducing the movement required to transport the pallets.

	
Amount of storage (space)	Assuming a pallet size of 1m x 1.2m with the racks being able to hold 15 in each row and having 3 levels and according to Error! Reference source not found. the storage area will not be larger than 20m x 20m for one day's worth of product. This only includes storage space and excludes manoeuvrability space. With one day buffer period considered, this space will double.
Holding period	Not Longer that 2 days after product was manufactured.
Component dependencies	Requires that pallets be transported to the storage area and be placed on the rack. The process also requires the pallets to be collected from the rack and transported to the loading areas.
Storage component movement	Can be transported with forklift or similar equipment.

6.12.Calculating Machine Requirements

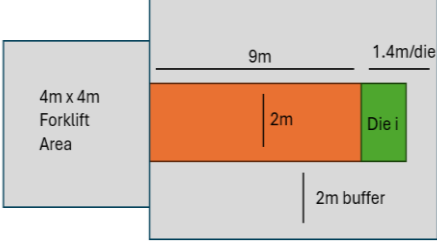
It is critical to calculate the machine requirements in terms of number, space, and supporting resource requirements. Each section described in the process flow diagram requires these parameters based on the user requirements. For the majority of processes, the demand will dictate the requirements and result in the resources required per process. The following equation describes the number of machines required:

$$F = \frac{SQ}{EHR}$$

The variable S describes the standard time the machine operates on a unit, and Q represents the number of units demanded per shift for the operation. Furthermore, the E gives the actual operating time on each unit and R the reliability of the machine. For these two parameters assumptions with support are often required to allow the variability in the process. Finally, H describes the available operating time for the machine. The result gives the number of machines required. More assumptions are required for operators and surface area requirements closely referenced to each process's data overview.

Table 17: Machines, space, operators requirements

Process	Number of machines	Space required	Operators needed
6.1 Inbounding Process of Round bars	Truckloads necessary per day: Super Link 6m + 12 m flat deck trailer 118 183kg ÷ 36 000kg (max legal Super Link trailer truck load) = 3.28 ≈ 4 Super Link Trailer trucks per day	Depending on design. One super link truck dimension (LxW) = 22m x 2.6m	2 operators per load
6.2 Round Bar Storage	Not applicable	<i>Total Volume</i> = <i>volume of one round bar</i> × <i>number of round bars</i> 16m ³ Needed for the round bar storage per day. 2m ³ Extra space for stacking shelves. <i>Total space required per day</i> = 18m ³	Not applicable
6.3 Welding and Coiling Process	S = 3s Q = 255587/day E = actual machine performance (from the overview and source) the data shows an actual welding time of 3s but double the changeover time. We can thus assume E = 50% R = 99% H = (8*60*60/3s = 9600) 9600 units per day/machine F = 16.15 ≈ 17 machines	17x5x10m for the operation = 850 m ² then 850 x 17% movement for operators = 995 sqm furthermore with 4m aisle along the coiling area to allow for forklift motion.	2 operators per machine and 1 forklift operator.
6.4 Round Bar Coiling and Storage	Each welding machine will have one coiling machine right after it that is operated by the two operators already present.	The space requirements for coiling are accounted for in the welding space. For the storage, each drawing machine will need a 1x2 m storage for coils which	Same forklift operator as the welding

		also serve as a staging area to be fed into the drawing process. Total: $43 \times 2 \text{ m}^2 = 86 \text{ sqm}$	process operator.
6.5 Wire Drawing Process	<u>Nr of Machines (setup = number of dies*3.5min)</u> 0.71mm Wire (8-die system) $F = \frac{0.05 * 2011981608}{(8 * 60 * 60 - 28 \times 60) \times 0.99 \times 0.8}$ <i>F = 22 machines</i> 0.9 mm Wire (8-die system) $F = \frac{0.05 \times 1252148060}{(8 * 60 * 60 - 28 \times 60) \times 0.99 \times 0.8}$ <i>F = 14 machines</i> 1.8 mm Wire (5-die system) $F = \frac{0.05(125214806 + 500859224)}{(8 * 60 * 60 - 17.5 \times 60) \times 0.99 \times 0.8}$ <i>F = 7 machines</i>	 0.71mm Wire $150 \text{ m}^2 \times 22 = 3300 \text{ m}^2$ 0.9mm Wire $150 \text{ m}^2 \times 14 = 2100 \text{ m}^2$ 1.8mm Wire $124 \text{ m}^2 \times 7 = 868 \text{ m}^2$	0.5 x operator/ Machine = 21 4 Forklift operators & 4 Forklifts
6.6 Scraping and Storage	Daily Weight of Scrap = $(6250 + 6250 + 2500 + 10000) / 220 \times 0.04 = 4.59$ tons per day Assuming one bin can contain one ton of scrap. Number of bins = 5.	Bin requires 1m x 1m worth of floor space with a clearance all-around of 0.5 m, this equates to 4 m^2 per bin. ($2\text{m} \times 2\text{m} = 4$) Total = $5 \times 4 = 20 \text{ m}^2$ for bins The forklift requires 4 meters width of manoeuvrable space along the backside of the bin. The length of the aisle depends on the path of the forklift.	Requires one forklift operator to empty bins.
6.7 Galvanising and Cooling	Daily tons of wire to be galvanised = $(6250 + 6250 + 2500 + 10000) / 220 \approx 114$ tons per day Assuming machines are capable of galvanising 30 tons of wire per day, equates to 4 machines.	The machine requires 10m x 105m of floor space. Including 0.5m of clearance all around, total space required per machine equates to 1166 m^2. ($11\text{m} \times 106\text{m} = 1166$)	Requires 2 operators per machine.

	(throughput rate of 30-45 tons per day, but chose 30 tons to leave some room for error) (114/30 ≈ 4)	Total = 4 x 1166 = 4664m ²																					
6.8 Zink Inbound Raw Materials	1 Forklift necessary 1 Interlink truck load every 13 days as shown in Table 11.	1 Forklift need an aisle width of 4m. <i>Storage space for pallets = 7m² for 13 days</i> Truck moving space depending on design.	Same operators as inbounding process of round bars.																				
6.9 Wire Spooling and Storage Process	Spooling is done as part of the galvanization process. For storage, we assume the weaving process and galvanization process and the weaving process is balanced. We will give assume a 50% daily demand buffer at the decoupling point.	The rolls take 1x1m/roll and stack to a height of 3. <table><thead><tr><th>Diameter</th><th>Daily Demand</th><th>Rollers/day</th><th>50% buffer</th></tr></thead><tbody><tr><td>0,71</td><td>9145370,95</td><td>28,57928</td><td>4</td></tr><tr><td>0,9</td><td>5691582,09</td><td>17,78619</td><td>9</td></tr><tr><td>1,8</td><td>569158,209</td><td>1,778619</td><td>1</td></tr><tr><td>1,8</td><td>2276632,84</td><td>7,114478</td><td>4</td></tr></tbody></table> Total: (1sqm x 28 /3) + 4m buffer zone for forklift movement	Diameter	Daily Demand	Rollers/day	50% buffer	0,71	9145370,95	28,57928	4	0,9	5691582,09	17,78619	9	1,8	569158,209	1,778619	1	1,8	2276632,84	7,114478	4	1 operator that is also the forklift driver.
Diameter	Daily Demand	Rollers/day	50% buffer																				
0,71	9145370,95	28,57928	4																				
0,9	5691582,09	17,78619	9																				
1,8	569158,209	1,778619	1																				
1,8	2276632,84	7,114478	4																				
6.10 Weaving Process	Using the calculations from Table B1.1 in Appendix B a total number of 105 machines are required.	Referring to the figure within the Process Space Requirements section in Table 13 Weaving Data Overviewthe space required will be 21m x 8m which equates to 168m ² per machine. This includes a 0.5m clearance all-around. Total = 105 x 168 = 17640m ²	Requires 2 operators per machine.																				
6.11 Packing and Packaging	<u>Strapping:</u> Using the calculations from Table B1.1 in Appendix B and the process throughput information from Table 14 the following calculations can be done. Rolls of wire to be strapped per day = 5620. Throughput rate = 30 seconds per roll.	<u>Strapping:</u> Machines are hand-held, however take 1.5m x 3m of work area per machine. This equates to 4.5m ² per machine. Total = 8 x 4.5 = 36m ² for machines. <u>Stretch Wrapping:</u> Machine requires 1.595m x 0.6m = 0.957m ² of floor space.	<u>Strapping:</u> Requires one operator per machine. <u>Stretch Wrapping:</u> One operator per forklift.																				

	$F = \frac{SQ}{EHR}$ $= \frac{30 \times 5620}{0.99 \times 0.75 \times (8 \times 60 \times 60)}$ $\approx 8 \text{ Machines}$ E was estimated at 0.75 to leave some room for error. <u>Stretch Wrapping:</u> Assuming a throughput rate of 20 pallets per hour as in Table 15. $F = \frac{SQ}{EHR}$ $= \frac{3 \times 352}{0.99 \times 0.85 \times (8 \times 60)}$ $\approx 3 \text{ Machines}$ The number off pallets per day are calculated in Error! Reference source not found. in the Appendix. Working on an average of 20 pallets per truck, assuming a 40 foot long truck, the average number of truck loads outbound equates to $352/20 \approx 18 \text{ truck loads}$	Including 0.5m of clearance all-around equates to $4.152m^2$ per machine. Total = $3 \times 4.152 \approx 13m^2$ for machines. Forklift requires 4 meters width of manoeuvrable space along the backside of the bin. The length of the aisle depends on the path of the forklift. <u>Storage:</u> From the Amount of storage section in Table 16 the required area will be 20m x 20m which equate to $400m^2$. The area required for the forklift depends on the design of the layout.	Assuming one forklift per machine <u>Storage:</u> Same operators and forklifts used for Stretch wrapping.
			Total: 292

Table 18: Additional Space Considerations

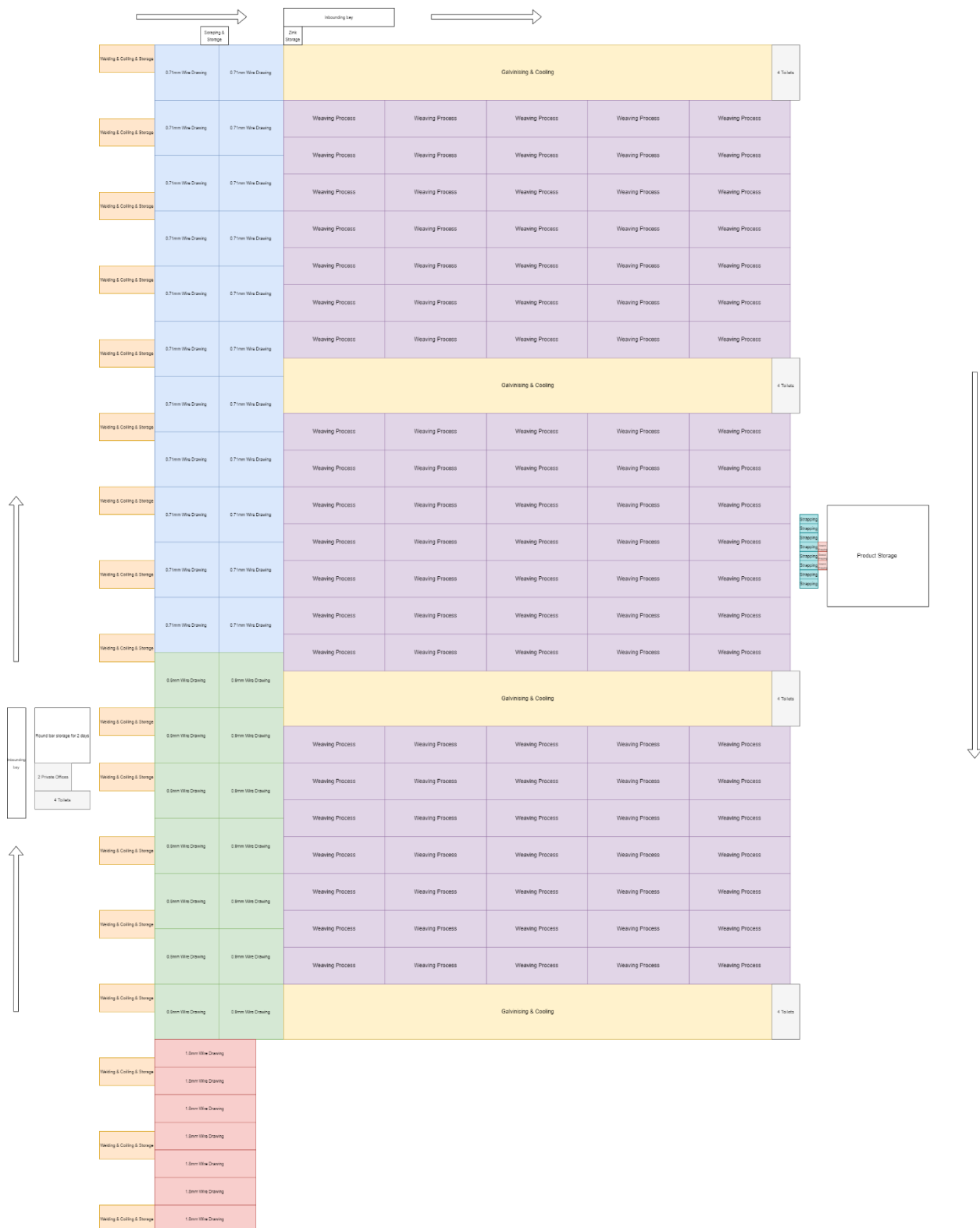
	Space required	Total Space
Restrooms	0.8m x 1.6m per toilet with extra 0.45mx0.8m for a sink Total of 2/3 sinks for total toilets and 20% extra moving space. 15 employees per toilet required. The ratio for operators to managers is 80:20.	292 operators 59 supervisors and managers 50% of the floorspace required for cleaning therefore $(50\,000 \times 50\%)/1100 = 23$ cleaners required A total of 374 people need restrooms.

	One cleaner per 8 hour shift can clean up to 1100 square meters.	Total restroom space required = $(375/15) = 25$ toilets = $25 \times (0.8 \times 1.6) = 32$ sqm $32 \times 2/3 = 21$ sinks required = $21 \times (0.45 \times 0.8) = 7.6$ sqm Total space with 20% moving space = $(32+7.6) \times 120\% = 48$ sqm
Offices	Open plan workstation approximately 5.5 sqm per person. Double story with offices on top and a kiosk and lounge on the ground floor.	$5.5 \text{ sqm} \times 59 = 325$ sqm
Parking	5mx2.5m per parking	$374 \times (5 \times 2.5) = 4675$ sqm Visitors parking = $20 \times (5 \times 2.5) = 250$ sqm Total parking space = 4925 sqm

7. Design alternatives.

Different designs were created, each offering a unique design ensuring flexibility and accommodating the different needs and requirements. The designs were designed on a software called Draw.io.

7.1. Alternative 1



7.1.1 Alternative 1 Critical Decisions

Alternative 1 is the first concept/alternative design. This design is a combination of product and process layout where similar equipment is mostly group together, but the focus is on a product layout where the equipment, tools, and machines are located according to how a product is made.

Inbounding Process and Storage of Round Bars Key Design Considerations

- The inbound process is designed so that the trucks can easily drive straight into the bay and then drive around the facility to exit on the other side and that no turning is required and that the process flow of deliveries are effective.
- Positioning the round bar storage near the inbound bay allows for easy material transfer by the crane.
- The strategic placement of toilets near the delivery point ensures easily accessible restroom facilities for truck drivers, minimizing disruptions during their deliveries.

Welding, Coiling and Storage Process Key Design Consideration

- Placing the welding, coiling, and storage areas between the drawing processes ensures smooth material flow from one stage to the next.

Drawing Process, Galvanising, and cooling

- The galvanising machines are located between the weaving machines, to reduce the distance materials need to travel.

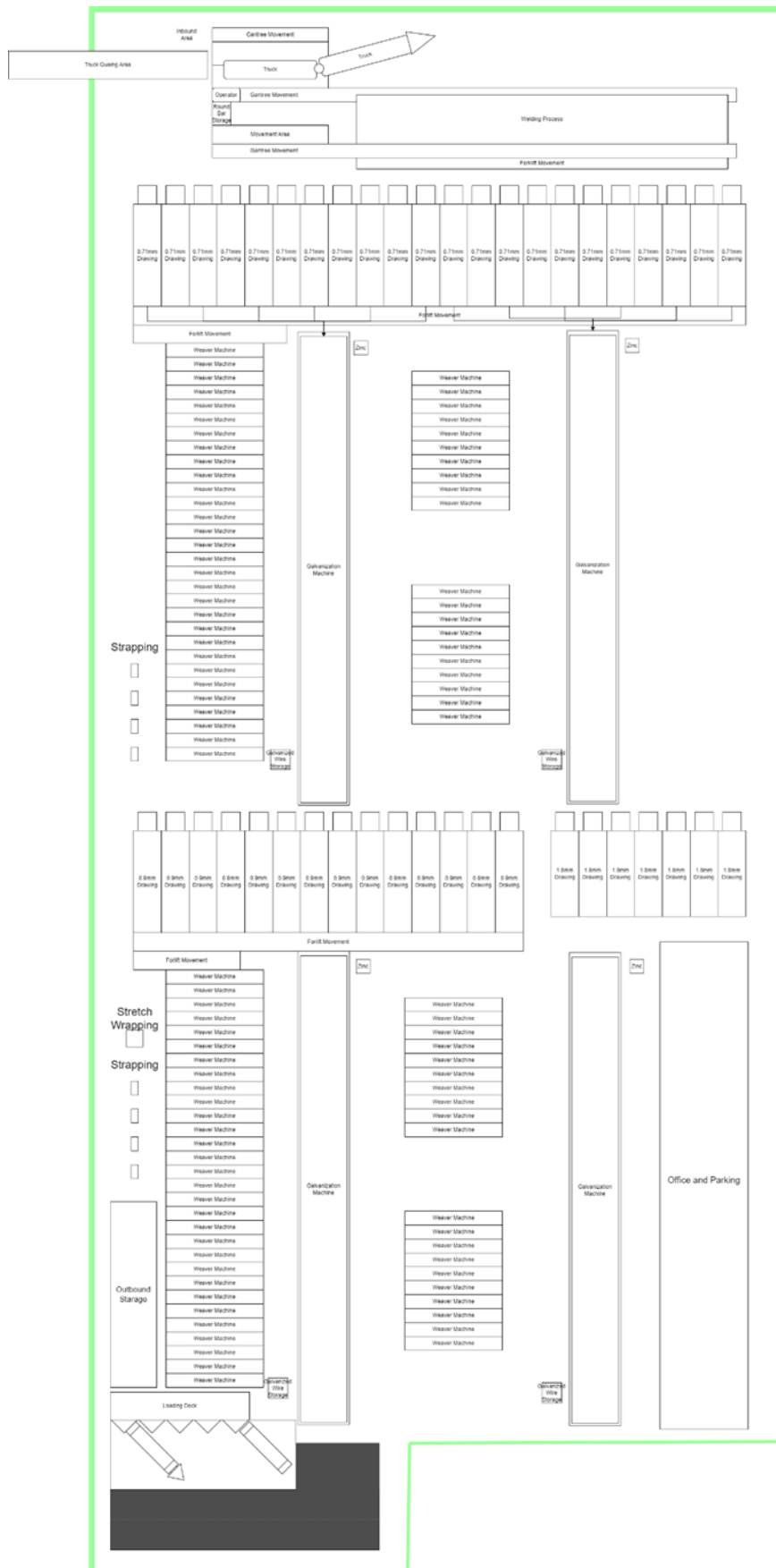
Strapping, Stretch Wrapping and Outbound Product Storage

- These machines are located in the end and middle of the facility so that the final product can be transferred to storage in a shorter distance.

Total Space Occupied

Alternative 1 occupies more or less 47 000 sqm.

7.2. Alternative 2



7.2.1 Alternative 2 Critical Decisions

Alternative 2 is the second concept/alternative design. The design is based on an assembly line approach that aims to achieve a smooth flow of materials. During the design process that follow is placed in the following order with some material handling process (overhead crane/gantry, forklift, etc.) linking them.

Loading Area Key Design Considerations

- The loading area aims to have easy access for trucks.
- Querying areas to avoid congestion.
- Proximity to storage

Welding Process Key Design Consideration

- Proximity to raw material storage
- Movement space for overhead gantry easy flow into the drawing process area
- Clear transportation to all drawing processes.

Drawing Process and Galvanization Key Features

- Staging Area and Work-In-Progress storage for coils
- Forklift Movement Area
- Close to Welding Process
- Direct Feed into galvanization
- Galvanization machine proximity to Storage

Strapping, Stretch Wrapping, Outbound Storage, and Outbound Logistics Area Key Features

- Proximity of three processes to minimize material flow.
- Outbound logistics has a staging area and 45-degree loading dock for 40ft trucks.

Total Space Occupied

Alternative 2 occupies 50 213 sqm which is equal to 5 hectares.

7.3. Alternative 3



7.3.1 Alternative 3 Critical Decisions

Alternative 3 is the third concept/alternative design. The design groups different process together in a linear fashion with similar machines placed next to one another.

Weaving Process

The weaving process and equipment have been placed in such a manner that the output can be transported by a conveyor system. This will allow all the rolls of wire to be transported to one location, where the packaging process will start.

Strapping, Wrapping, and Storage

The location for the strapping of the rolls, the wrapping of the pallets, and the outbound storage have been strategically placed together. The rolls of wire all move to this area before it is stored and then delivered/collected. This allows for centralised strapping, wrapping and storage.

Inbound Area

The design incorporates an inbound delivery bay that allows large trucks to drive in where an overhead gantry can be used to unload the material.

Outbound Area

The design incorporates an outbound dock where trucks can be loaded with ease. This area includes docks for delivery trucks to load as well as a dock where collections can be done.

Positioning

- The inbound and outbound area was deliberately placed on opposite sides of the facility and away from parking to avoid digestion of trucks.
- The offices and parking were placed along the front side of the facility to allow easy access to offices and parking, it will also be easy to access the production floor from the offices and will give a clear view of the warehouse floor.

Total Space Occupied

Alternative 3 occupies $\pm 58\,000\text{ m}^2$ which is equal to 5.8 hectares.

8. Evaluating alternatives

For the evaluation of the different alternatives, an acceptance criterion has been set up to ensure that all the requirements are met. The evaluation is done in Table 19.

Table 19: Evaluation of alternatives

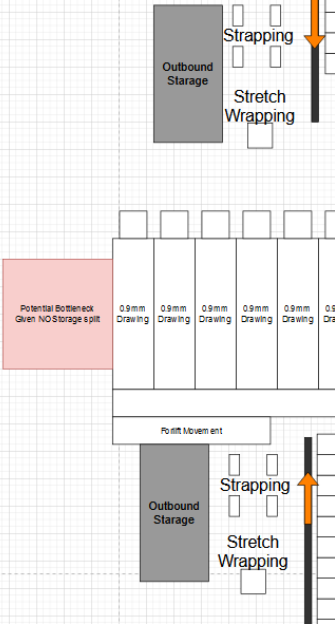
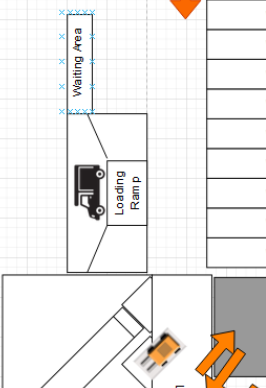
Requirements and criteria	Alternative 1	Alternative 2	Alternative 3	Best Design Option
Traveling distance of material (Low/Medium/High)	High	Low	Medium	Alt 2
Space usage	47 000 sqm	50 000 sqm	58 000 sqm	Alt 1
Activity relationship (Low/Medium/High)	Medium/High	High	Medium	Alt 2
Smooth process flow (Yes/No)	No	Yes	Yes	Alt 2 / Alt 3
Congestions or delays (High/Low)	High	High	Low	Alt 3
Handling equipment (Good/Bad)	Good	Good	Good	-
Allocating available space (Good/Bad)	Very Good	Bad	Good	Alt 1

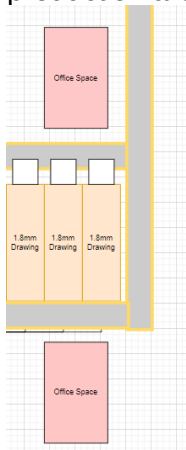
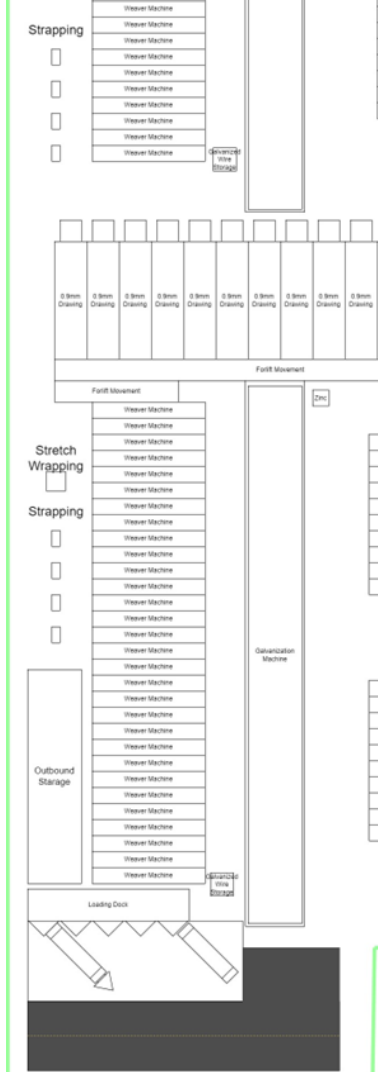
Flexible for future expansion (Low/Medium/High)	High	High	Low	Alt 1 / Alt 2
Within 100Ha space (Yes/No)	Yes	Yes	Yes	-
Included parking (Yes/No)	No	Yes	Yes	Alt 2 / Alt 3
Included rest rooms (Yes/No)	Yes	No	No	Alt 1
Included offices (Yes/No)	Yes	Yes	Yes	-
Included sufficient space for delivery lot (Yes/No)	Yes	Yes	Yes	-
Ordered from best design considering all the requirements and criteria: Counted as the number of times that the design was the best in each criteria.				Alt 2 = 5 Alt 1 = 4 Alt 3 = 3

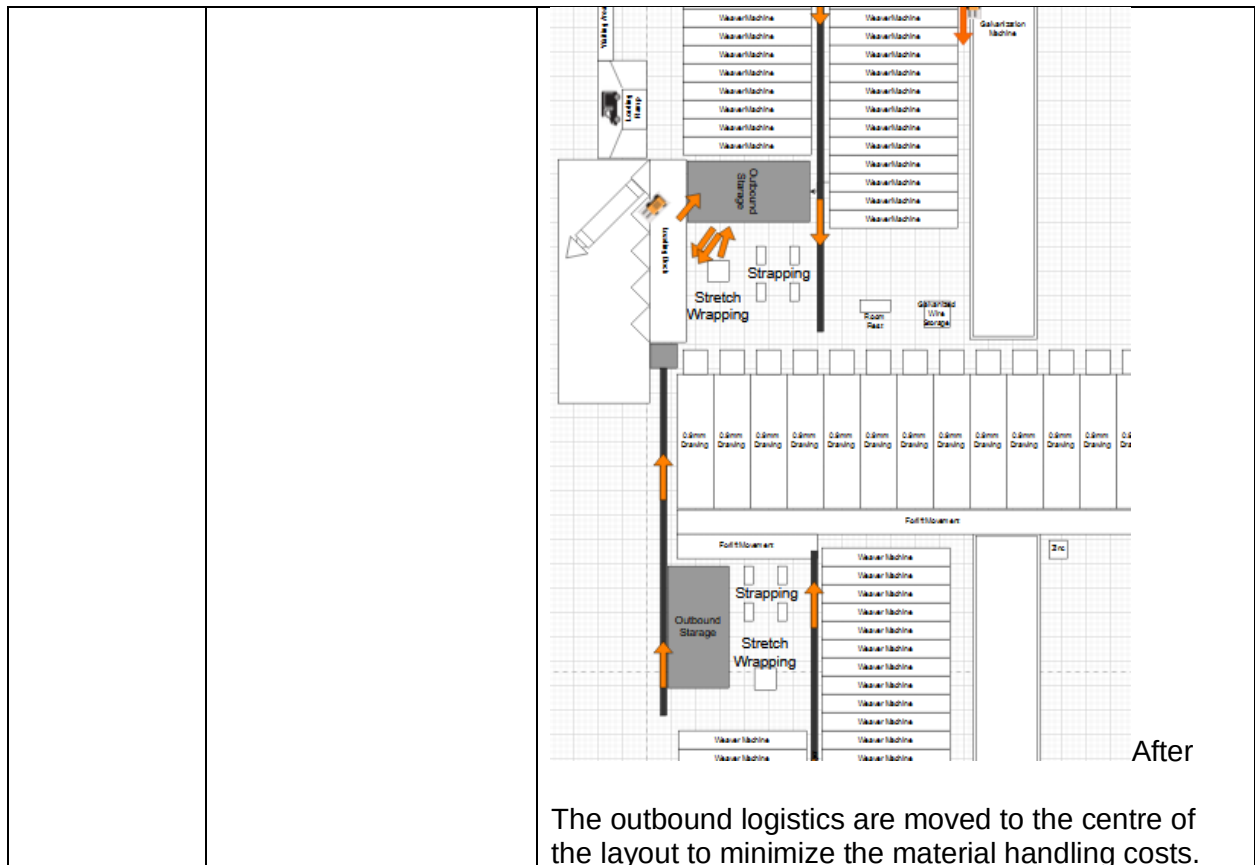
In the comprehensive analysis, three alternative designs for the facility were examined. The final design will integrate components and parts from all three options, thereby combining the optimal attributes of each to align with the predetermined requirements and evaluation criteria.

9. The final design

Table 20: Changes for final design

Number	Process	Action and Reasoning
1.	Truck inbounding	From Alternative keeping the truck's entry straight reduces the need for manoeuvres and it also saves time for the delivering. A designated one-way route for trucks helps avoid congestion, ensuring smooth traffic flow and efficient logistics operations.
2.	Galvanization machines and Weaving	Move the galvanization Process on alternative 2 to look more like Alternative 3. The changes remove the need to travel around the galvanizing machines to storage and distribution. The move facilitates a better flow of material and the introduction of a conveyor for weaved netting.
3.	Strapping, wrapping, and Outbound Storage	Move strapping and wrapping to centre to converge transport and material flow. Split storage for area outbound in two to reduce transport through bottleneck areas visible in the diagram. 
4.	Small Collection Area	An additional small order collection area with two vehicle waiting area is introduced to the design. 

5.	Office	The Office Space was split into two indicated in red, so that the offices will have a clear view of the production area. 
6.	Outbound Logistics	 Before



9.1. Final Layout Design

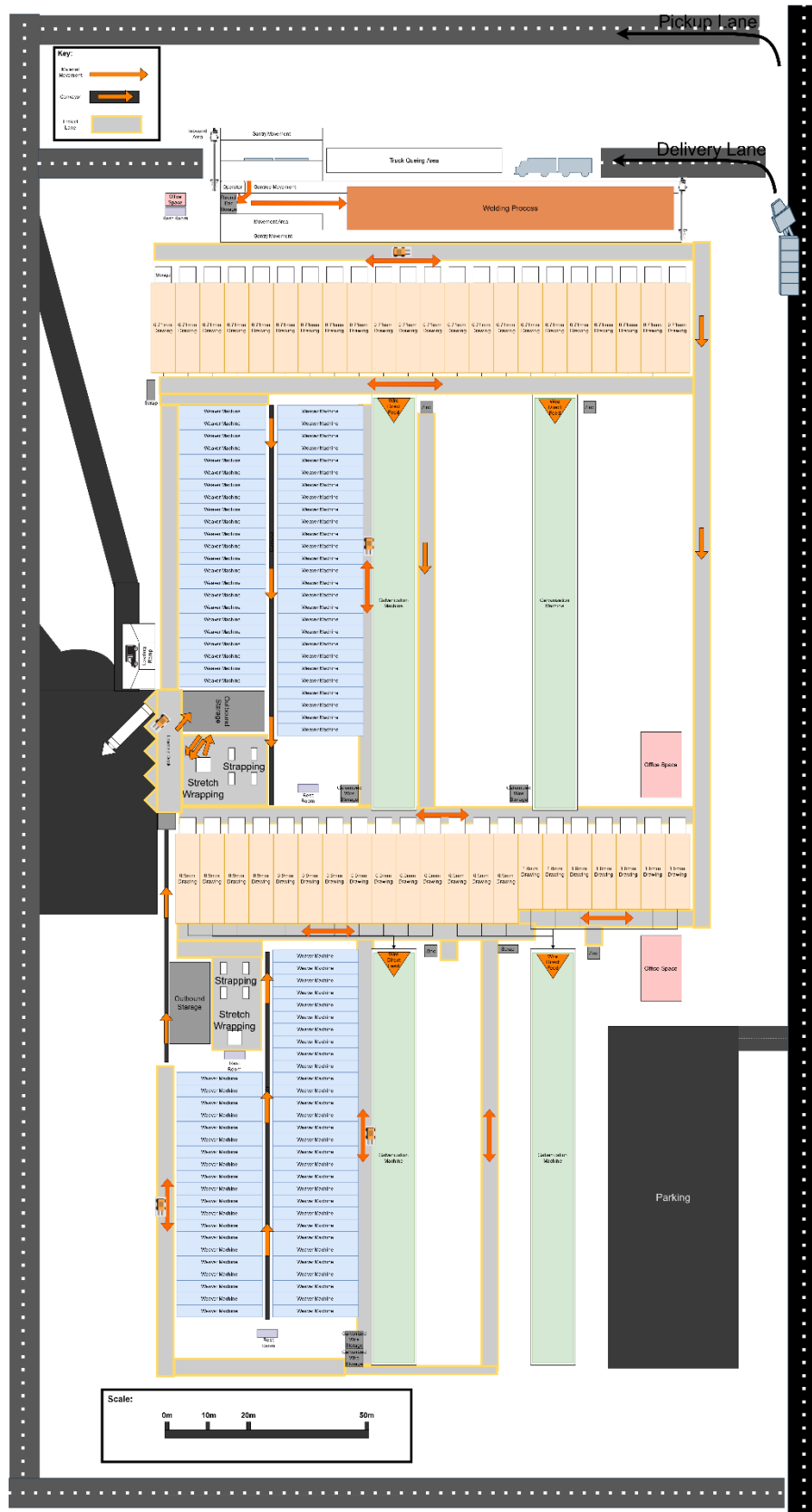


Figure 13: Final Facility Design

The final facility design contains some key features and concept that originated from the alternative designs as well as some improvements. These features include the concept of using a conveyer system to move the items to a designated area for next operations. The conveyer was placed in the centre of two columns of machines for minimal conveyer equipment needed. Furthermore, the rolls of wire are being moved by the conveyor system to the centre of the facility, to centralise the outbound loading area and to be able to do the strapping, wrapping and storage in an area near each other.

The design features a dedicated inbound unloading bay. Trucks move in under this bay where an overhead gantry is available to unload the truck quickly and with minimal effort. This bay has a restroom in proximity which the truck drivers can use while the truck is being unloaded. The facility includes an outbound loading area, this loading area comprise of loading docks – positioned at an angle for ease of loading – and ample space for trucks to manoeuvre. This outbound area incorporates a loading ramp for small order collections and are not in the way of trucks being loaded.

Forklift lanes are utilised to reduce the risk of injury of operators in the facility. Furthermore, the facility has two office spaces placed near the centre of the facility, giving a clear view and easy access to the production floor. The parking area is placed next to one of these office spaces to save some space and reduce the distance between the office and the parking. In addition, the facility design left some room for future expansion to meet the need for possible growth in demand.

The introduction of roads that flow in a continuous approach around the facility aids movement of raw materials and the collection of complete products. The inbound lot allows to unload with minimal maundering given the drive through approach. The road that circles the facility reduces risk of traffic congestion and reduces intrusion on surrounding infrastructure.

Given the design and considerations made the requirements of the client is met. The design allows the implementation of construction to commence. The design will facilitate the demand and production of products while maintaining steady flow of materials without high handling costs. Space is well utilized while leaving room for expansion. All the necessary detail is included to lead to implementation of the design.

9.2. Relative Sizing of Design

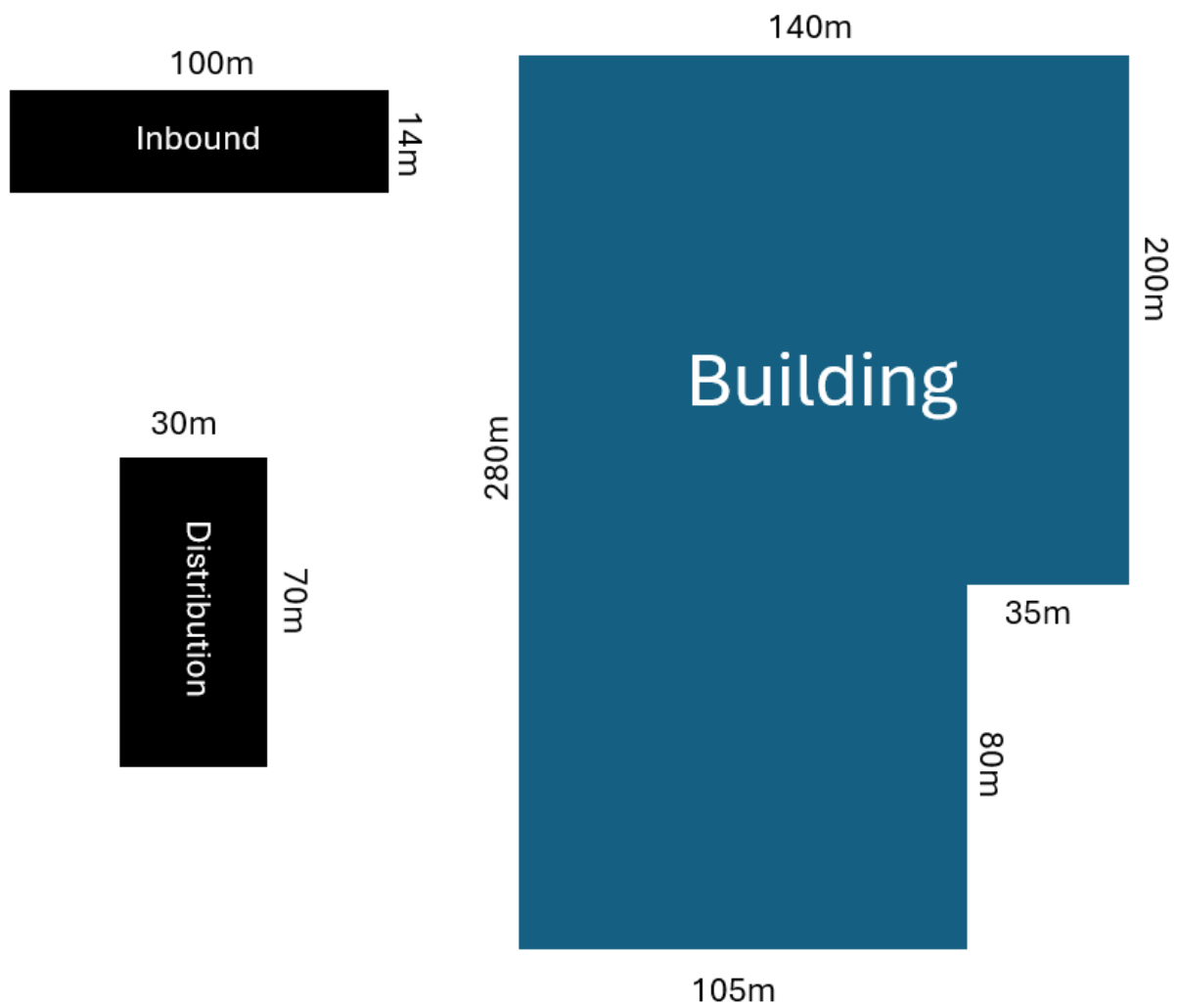


Figure 14 Relative Parameter Sizes

9.3. Design Features

Table 21: Key Attributes

Feature	Attribute
Building Excluding Roads, Parking, and Lots	38 800 sqm
Total Green Field Land Required roads inclusive.	56 727 sqm
Roads and parking lots	11 247 sqm

Office Coverage	325 sqm
Restroom Coverage	48 sqm
Real Estate to Acquire	72 000 sqm

10. Business case

Table 22: Cost for equipment

Cost for Equipment		
Type	Cost	Total cost
Overhead gantries	Capacity = 10t Span = 10.2-31.5m Lifting height = 16m Price = R472 150	$R\ 472\ 150 \times 2 = R944\ 300$
Round bar storage racks	To accommodate at least 150 000kg = R760 000	R760 000
Butt Welding Machines	R 55 000 per butt welder	$R\ 55\ 000 \times 17 = R935\ 000$
Coiling Machine	N.A. Price included in Drawing machine, since machine Coil wire after drawn.	N.A.
Drawing Machine	R200 000 per Machine	$R200\ 000 \times (22 + 14 + 7) =$ R8 600 000
Galvanising Machine	R 5 700 000 per machine	R22 800 000
Weaving Machines	R1 615 000 per Machine	$R1\ 615\ 00 \times 105$ $= R169\ 575\ 000$
Strapping	R10 400 per handheld electric strapping machine.	$R10\ 400 \times 8 = R83\ 200$
Stretch Wrapping Machine	R100 000 per Machine	$3 \times R100\ 000 = R300\ 000$

Storage racks	Storage racks that are 3 levels high and can take the final products' weight: R760 000	R760 000
Conveyor Belt	R40 000 per Meter	R40 000 x (98+90+57) = R9 800 000
Forklifts	R349 995	R349 995 x 11 = R3 849 945
Scrapping Bins	R8 550 per bin	R 8 550 x 5 = R42 750
Total		R218 450 195

Table 23: Cost for building etc.

Costs for building etc.			
Component	Rate [R/sqm]	sqm/Phase	Costs
Industrial Warehouse Brick Walls, Steel Frame, Ceiling, Roof Sheets	R 6 900,00	38427	R 265 146 300,00
Offices	R 10 000,00	325	R 3 250 000,00
Parking and Roads	R 800,00	11 250	R 9 000 000,00
Restrooms	R 7 500,00	48	R 360 000,00
Office Electrical Including Electronics	R 1 700,00	325	R 552 500,00
Factory Electricity and Electronics	R 1 300,00	38800	R 50 440 000,00
Fire Safety	R 275,00	38800	R 10 670 000,00
Air-Conditioning	R 1 200,00	373	R 447 600,00
Industrial Real Estate	R 7 000,00	72000	R 504 000 000,00
Total:			R 843 866 400,00

10.1.Capex

Capex accounts for the capital costs that are associated with the initiation of the project. Capital expenses are the money that must be sunk into the project. Capital can be raised through loans or from cash that wire.com has in reserve to invest in the project.

Table 24 Capital Expense of Project

Capital Expense	
Equipment	R 237 000 000
Construction	R 340 000 000
Real estate Acquisition	R 504 000 000
30 % Contingency Factor	R324 300 000
Total Capex	R 1 405 300 000
Loan Payment at 7.5% and 20 Years [Monthly]	R 74 970 000

10.2.Operational Costs

Table 25: Labour cost

Operational Costs (per Year)		
Managers	59 supervisors and managers	R 541 204 x 59 = R31 931 036
Operators	292 operators	R366 292.2 x 292 = R106 957 410
Cleaners	23 cleaners	R161950 x 23 = R3 724 850 per year
Total:		R 142 613 296 / year

Table 26: Electrical costs

Machine	Power Consumption/machine	Number of Machines	Total Consumption/Years
Butt Welder	1 kWh	17	29 920 kWh
Wire Coiling	3 kWh	17	89 760 kWh
8 die Drawing	176 kWh	36	11 151 360 kWh
5-die Drawing	110 kWh	7	1 355 200 kWh
Galvanization	50 kWh	4	352 000 kWh
Weavers	22 kWh	105	4 065 600 kWh
Lights	0.1 kWh/sqm	38800	6 828 800 kWh
Uncertainty	30%		
Total			31 034 432 kWh
Cost [R1.45/kwh]			R 44 999 927/year

Table 27 Raw Material Cost

Material	Cost	Total
Galvanisation Zinc	R40 000 per Ton = R40 per Kg	R40 x 3680kg = R147 200 per day = R32 384 000 per year
Mild Steel	R42.40 per 3m coil (Diameter = 5mm)	R42.40 x 255 587units = R10 836 889 per day = R2 384 115 580 per year
Total:		R 2 416 499 580

Table 28: Cleaning Supplies costs

Cleaning Supplies	
Only 50% of the total area can be cleaned and therefore 38800 sqm x 50% = 19 400sqm are used for the calculations regarding cleaning.	19 400sqm x R180 = R3 492 000 (Gambao, E., Hernando, M., & Surdilovic, D., 2008. Development of a Semi-Automated Cost- Effective Facade Cleaning System. https://doi.org/10.5772/6190 .)
Total Inc 30% Contingency	R 4 539 600

10.3.OPEX on Facility

Table 29: Operational Cost of facility

Operational Costs of Facility	
Raw Materials	R 2 416 499 580
Payroll	R 142 613 296
Consumable Supplies	R 4 539 600
Electricity Costs	R 45 000 000
Total [Year]	R 2 608 652 476

10.4.Risk Assessment

Different risks associated with the manufacture of this wire netting facility were identified, evaluated and mitigations are set in place to ensure the success of the facility.

Risks:

1. Not capable of meeting the demand with the current facility.
2. Risk for high traveling distances of the material.
3. Risk for not having enough storage for raw material and final products.
4. Risks for bottlenecks.

5. Workplace safety hazards.
6. Equipment failure.
7. Fire and explosion risks.
8. Traffic congestion of forklifts and trucks.
9. Environmental impacts leading to delays.
10. Market fluctuation.
11. Quality control.
12. Loadshedding.

Risk Matrix		Impact				
		Insignificant	Minor	Moderate	Major	Severe
Likelihood	Almost Certain	Medium	High	Very High 12	Very High 1	Very High
	Likely	Medium 5	High 2	High 4	Very High 3	Very High 6
	Possible	Low	Medium 8	High 11	High 10	Very High
	Unlikely	Low	Low	Medium	Medium 9	High
	Rare	Low	Low	Low	Low	Medium 7

Figure 15: Risk Matrix Before Mitigations

Mitigations:

1. Room for expansion in the facility layout.
2. The use of the conveyor belt.
3. Room for expansion for storage.
4. Different workstations and buffers.
5. Immediate stop buttons on equipment. Forklift lanes to not let employees are in the way of forklifts.
6. Regular maintenance.
7. Fire emergency strategy in place.
8. Traveling lanes for forklift implemented and truck roads are only one-ways with dedicated loading areas.

9. Safety stock with enough storage space.
10. Make use of appropriate forecasting methods and flexible production.
11. Manages and supervisors to ensure and test quality throughout the process.
12. Enough space for backup power supply.

Risk Matrix		Impact				
		Insignificant	Minor	Moderate	Major	Severe
Likelihood	Almost Certain	Medium	High	Very High	Very High	Very High
	Likely	Medium	High	High	Very High	Very High
	Possible	Low	Medium	High 12	High 1	Very High
	Unlikely	Low	Low 5 2	Medium 11 4	Medium 3 10	High 6
	Rare	Low	Low	Low	Low 9	Medium 7

Figure 16: Risk Matrix after Mitigations

10.5.Project Schedule

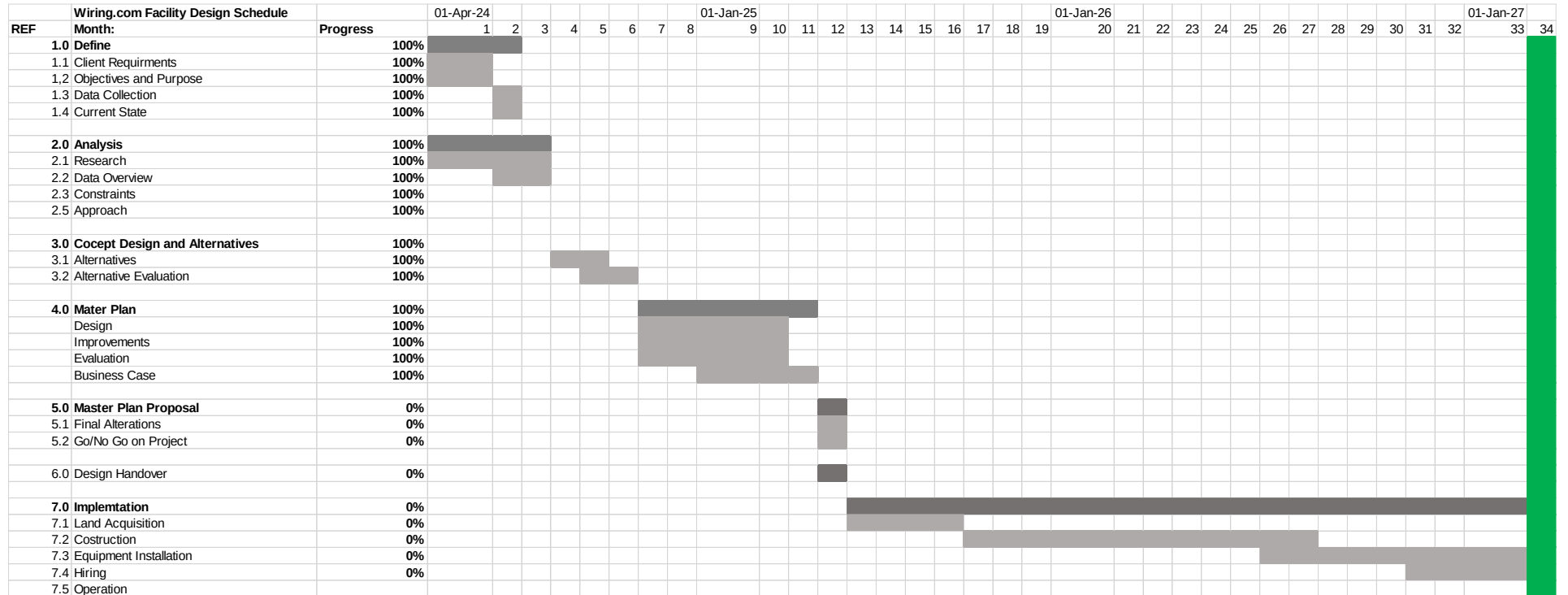


Figure 17 Project Schedule

APPENDIX A: EQUIPMENT

A1 Wire Drawing Process

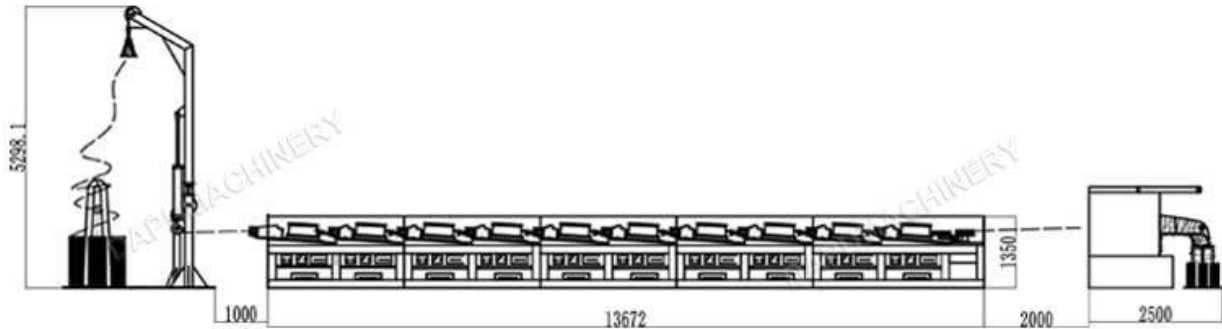


Figure A1: Wire drawing machine

(“Wire Drawing Machine | DAPU Drawing Machine Manufacturer”, 2019)

Model	LZ-560
Raw material	SAE1015, SAE1008, SAE1006 etc
Number of drawing pots	Max.12
Inlet wire diameter	8mm, 6.5mm, 6mm, 5.5mm etc
Outlet wire diameter	4mm, 3mm, 2.2mm, 2mm, 1.8mm, 1.6mm, 1.4mm etc
Each motor capacity	22KW
Drawing speed	Max.20m/second
Each pot weight	1500KG

Material that can be drawn:

Steel wire (High, middle, low carbon steel wire, stainless steel wire, pre-tension steel wire, bead wire, rubber tube wire, spring steel wire, cord wire, and so on).Welding wire (Air protecting welding wire, submerged arc welding wire, flux-cored wire, and so on).lectric wire and cable (Aluminum-clad steel wire, copper wire, aluminum wire, and so on).Alloy wire and other kinds of metal wire.

Figure A2: Wire drawing machine specifications.

(“Wire Drawing Machine | DAPU Drawing Machine Manufacturer”, 2019)

To calculate setup time two sources are conducted to calculate the number of dies required for each wire and the setup time per die is multiplied per die required. According to J. Sheu et al. The optimum combination of drawing die design using 20% reduction ratio. The setup time is taken from a video available online at https://www.youtube.com/watch?v=nrgv_wpeZQE. The average is taken to be 3min and 30 seconds per die present.

A2 Galvanisation Machine

Main parameter:
1. Wire diameter: 1.5mm-5mm
2. Output: 30-45 tons/DAY
3. Max. coil weight: 150 kg, the inner coil diameter: 40/50mm
4. Workhouse Size(W*L): 10m*105m
5. Energy: natural gas with 300mpa for outlet
6. Zinc coating: 70---300g/m2
7. Tensile Strength: 350~450N/mm, low carbon iron wire
8. Zinc pool life : 2 years with 450°C
9. Anneal Furnace Weight : 70 tons
10. Zinc furnace weight: 25 tons
11. Power: 50KW

Figure A3: Galvanisation machine specifications

Source: <http://www.china-anbermachine.com/wire-galvanizing-line/hot-dipped-wire-galvanizing-line.htm>

A3 Round Bar Welding

Sources:

<https://www.techweld.in/product-page/20-kva-pedal-operated-heavy-duty-wire-butt-welder>

<https://www.youtube.com/watch?v=GTgEpDk4V8k>

A4 Weaving Process

Supplier 1:



Figure A4.0: Weaving machine

Product Attributes

Model No.	: 1/2" 3/4" 1"
Brand	: HONGLIN
Wire Diameter	: Normally 0.45-2.2mm
Mesh Width	: Normally 2600mm,3000mm,3300mm,4000mm,4300mm
Motor	: 2.3kw(slat shaft motor)+5.7kw(twisted motor)+2.3kw(feeding motor)+2.3kw(rolling motor)
Note	: 1.One set machine can only do one mesh opening. 2.We accept an special orders from any clients.
Raw Material	: Galvanized wire , PVC coated wire
Mesh Size	: 1/2"(15mm);1"(25mm or 28mm);2"(50mm);3"(75mm or 80mm)
Working Speed	: If your mesh size is 1/2",it is about 60-65M/h If your mesh size is 1",it is about 95-100/h

Figure A4.1: Weaving machine specifications

Source: HEXAGONAL WIRE NETTING MACHINE China Manufacturer (wirenetting-factory.com)

Supplier 2:



Figure A4.2: Weaving machine (2)

Table A4: Weaving machine specifications (2)

Mesh Size(mm)	Max. Netting(mm)	of Weaving wire (mm)	Max.Dia of Bordering wire(mm)	Max Weaving- vein No.	Power(kw)	Real Production Capacity	Exterior Size(m) L×W×H	Weight(T)
60×80	4500	2.7	4	3/5	30	400-400 Sq.m /Hour	6.8×1.4×2.4	13
	5500	2.7	4	3/5	30	500-550 Sq.m /Hour	7.8×1.4×2.4	14
80×100	4500	3	4	3/5	30	480-520 Sq.m /Hour	6.8×1.4×2.4	13
	5500	3	4	3/5	30	480-520 Sq.m /Hour	7.8×1.4×2.4	14
100×120	4500	3.5	4	3/5	30	560-600 Sq.m /Hour	6.8×1.4×2.4	13
	5500	3.5	4	3/5	30	560-600 Sq.m /Hour	7.8×1.4×2.4	14

Source: Hexagonal Wire Mesh/Netting Machine, Heavy Duty Hexagonal Wire Mesh Machine | Metal Tech (wiremachinecn.com)

Supplier 3:



Figure A4.3: Weaving machine (3)

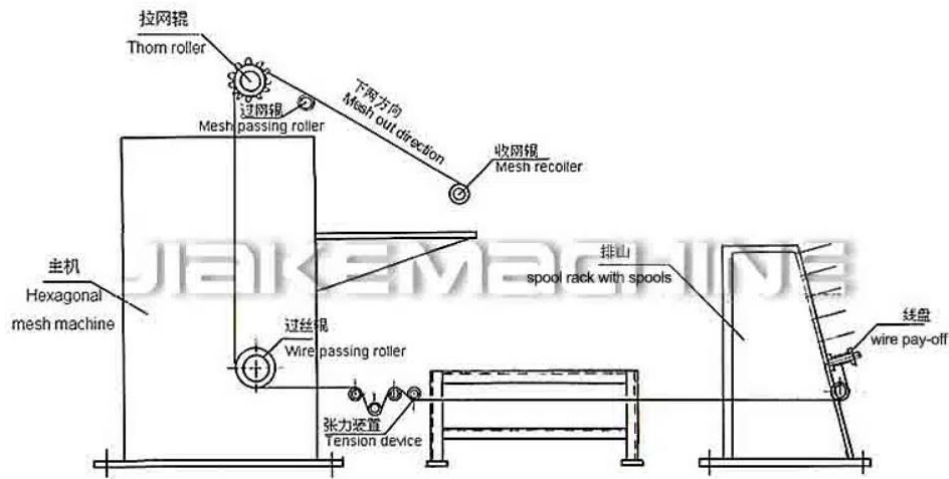


Figure A4.4: Weaving machine (3)

Table A4.1: Weaving machine specifications (3)

Speed	Mesh width	Mesh size	Wire diameter	Number of twists	Weight	Motor
60-150m/h	1200-3200 mm	12mm	0.38-0.5mm	3	2-2.5T	2.2kw
		15mm	0.40-0.70mm			
		18mm	0.38-0.70mm			
		20-22mm	0.38-0.80mm			
		25-27mm	0.50-1.00mm			
		30-32mm	0.50-1.00mm			
		40-44mm	0.40-1.30mm			
		50mm	0.50-1.50mm			
		75mm	0.50-1.65mm			
		100mm	0.50-1.80mm			

A5 Packing and Packaging Process

Table A5: Stretch wrapping specification

MODEL	GP1000L
Wrap size (L×W×H)	400-1200) × (180-400) × (300-600) mm
Machine Size (L×W×H)	1595×600×1194mm
Power Voltage	220 VAC
Total Weight	200kg
Wrap Efficiency	20-40 Pallets/Hour
Film Carriage	Power pre-Stretch system up to 350% with adjustable speed

Source: PALLET WRAPPERS - Glenpak Packaging and Machines



Figure A5.1: Wrapping machine



Figure 5.2: Packaging

Appendix B: Demand and Throughput Data

Table B1.1: Throughput and demand

CUSTOMER REQUIREMENTS	BIRD NETTING	CHICKEN NETTING	JACKAL NETTING	PIG NETTING
TON/YEAR	6 250	6 250	2 500	10 000
ROLLS/YEAR	277 778	149 165	128 206	680 723
ROLLS/DAY	1263	679	583	3095
ROLL/HOUR	158	85	73	387
METERS/YEAR	13 888 900	7 458 250	6 410 300	20 421 690
METERS/DAY	63 132	33 902	29 138	92 826
METERS/HOUR	7 892	4238	3 643	11 604
WEAVING MACHINE THROUGHPUT RATE	100M/HOUR	100M/HOUR	100M/HOUR	100M/HOUR
ESTIMATED NUMBER OF WEAVING MACHINES	(3 Rolls per Machine) 27 Machines	(1 Roll per Machine) 43 Machines	(5 Rolls per Machine) 8 Machine	(3 Rolls per Machine) 27 Machines

Table B1.2: Wire demand

NETTING TYPE	CUSTOMER DEMAND IN KG	RAW WIRE RADIUS [MM]	WIRE DENSITY OF MILD STEEL [KG/M ³]	LENGTH OF THE DIFFERENT DIAMETER WIRE PER YEAR [M]
BIRD	6250000	0,355	7850	2011981608

CHICKEN	6250000	0,45	7850	1252148060
JACKAL	2500000	0,9	7850	125214806
PIG	10000000	0,9	7850	500859224

CALCULATIONS:

To calculate the wire input of a specific diameter needed to create one meter of mesh for a specific netting product the following calculations have been performed:

Convert Customer requirements tons per annum to required weight in kilograms year:

This is done by multiplying the required tons per annum by a thousand to get required kilograms per year.

Converting required weight per annum to required meters of netting product per annum:

The weight in kilograms is divided by the respective weight per roll in kilogram, this gives the required rolls per annum for a specific netting product. Then multiplying this number of rolls per annum with the respective length per roll will yield the required meters of netting product per annum.

Obtaining the weight in kilograms of one meter of netting product:

Dividing the required weight per annum in kilograms with the required meters per annum of netting product yields the weight in kilograms per meter of netting product.

Obtaining the required input length of wire to produce one meter of a specific netting product:

Assuming a wire is cylindrical and have uniform mass the weight of a wire can be calculated by $W = [\pi \cdot r^2 \cdot h] \cdot \rho$, where W is the weight in kilogram, π is 3.14, r is the radius of the wire, h is the length of the wire, and ρ is the density of the material. In this case the material is mild steel, and the density will be $7850 \text{ kg} / \text{m}^3$.

Substituting the weight per meter of netting product into W , 3.14 into π , the radius of the specific wire into r , 7850 into ρ , and solving for h , the required length of wire in meter of that specific diameter wire necessary to produce one meter of the relevant netting product are obtained.

B2 Zinc Demand

Table B2.1: Zinc demand

Netting Type	Customer Daily Demand [m]	Raw Wire Radius [mm]	Surface Area [m ²]	Galvanized Coating [kg/m ²]	Zinc Demand [kg]
Bird	9145371	0,355	20389	0,07	1427
Chicken	5691582	0,45	16084	0,07	1126
Jackal	569158	0,9	3217	0,07	225
Pig	2276633	0,9	12868	0,07	901
				Total:	3679

B3 Zinc Transport

SHG Slab Ingot



Dimensions (Bundle)

Height	445 mm
Length	880 mm
Width	440 mm
Weight	1000kg +/- 50 kg

Composition

Zn: 99.995% min.	
Pb: 0.003% max.	Cu: 0.001% max.
Fe: 0.002% max.	Sn: 0.001% max.
Cd: 0.003% max.	Al: 0.001% max.

Figure 18: Zinc transport

("Zinc Metals | Sun Metals", s.a.)

Appendix C: Material Handling

Truck Payload

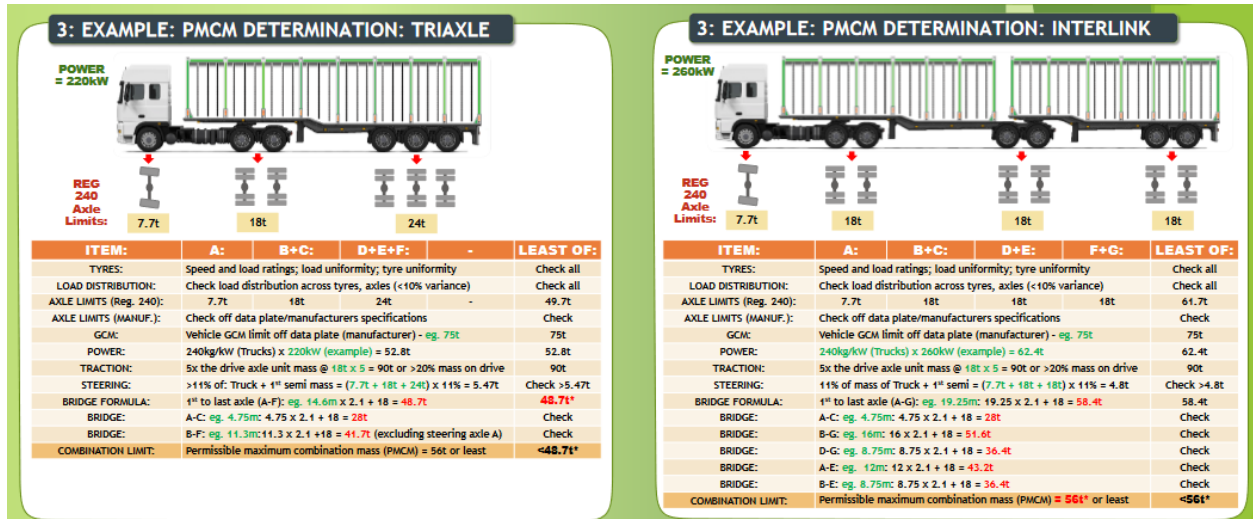


Figure C1: Truck payload

(SASARI, 2022)

Truck Size vs Pallet Carrying Capacity

Table C1: Truck size vs carrying capacity

Truck or Trailer size	Number of Pallets (48" x 40")
how many pallets fit in a 53 foot trailer	30 pallets
how many pallets fit on a 48 foot trailer	24 pallets
how many pallets fit on a 45 foot cube container	24 – 26 pallets
how many pallets fit on a 40 foot container	20 pallets
how many pallets fit in a 24 foot truck	12 pallets
how many pallets can fit in a 16 foot box truck	6 pallets

(griffin, 2014)

Shipping Area Maundering Allowance and Space Requirements

Table C1: Shipping area allowance

<i>Minimum Maneuvering Allowances for Receiving and Shipping Areas</i>	
Material Handling Equipment Utilized	Minimum Maneuvering Allowance (feet)
Tractor	14
Platform truck	12
Forklift	12
Narrow-aisle truck	10
Handlift (jack)	8
Four-wheel hand truck	8
Two-wheel hand truck	6
Manual	5

(Tompkins *et al.*, 2010)

Table 30: Space requirement

Truck Length (feet)	Dock Width (feet)	Apron Depth (feet)
40	10	46
	12	43
	14	39
45	10	52
	12	49
	14	46
50	10	60
	12	57
	14	54
55	10	65
	12	63
	14	58
60	10	72
	12	63
	14	60

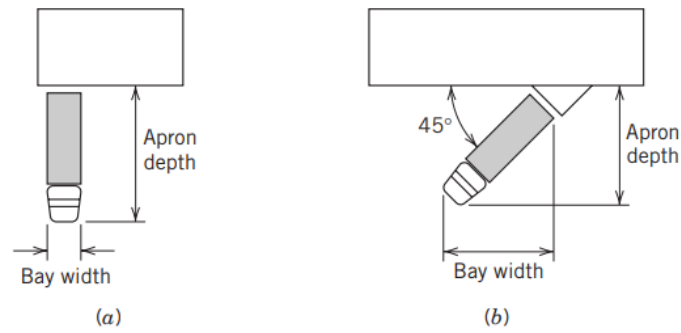


Figure 19: Turning truck

(Tompkins *et al.*, 2010)

Step	Description	Crew Size	USDA Reference Table Number	Productive Labor (Hours)
1	Spot carrier.	1	XII	.1666
2	Open truck door.	1	XII	.0163
3	Remove bracing.	1	XII	.1101
4	Get bridge plate.	1	XII	.0310
5	Position bridge plate.	1	XII	.0168
6	Place empty pallet at truck tailgate. Repeat 40 times.	1	XX	.2333
7	Unload cartons onto pallet. Repeat 640 times.	1	XIV A	1.5360
8	Pick up loaded pallet and move out of truck. Repeat 40 times.	1	VI	.3800
9	Transport pallet to storage area (100 ft) and return to truck. Repeat 40 times.	1	XI	.6040
10	Store pallet. Repeat 40 times.	1	VI	.1760
11	Remove bridge plate.	1	XII	.0048
12	Close truck door and dispatch truck.	1	XII	.0068
Total				<u>3.2817 hr</u>

Figure 20: Truck loading

Table 31: Storage Data

Total number of Rolls required per day	5620
Rolls per Pallet	Assume Max of 16
Total Pallets per day	352

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